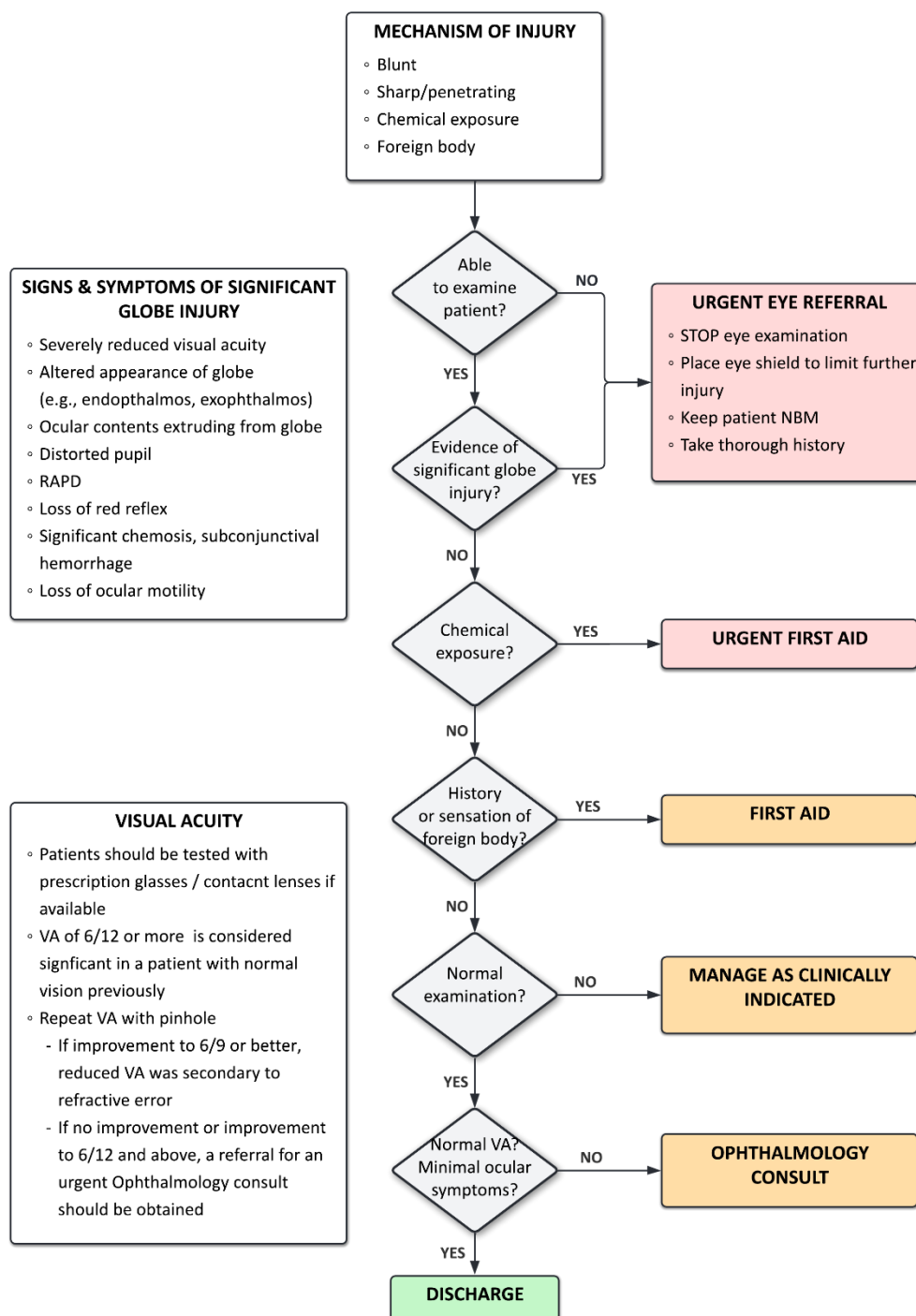


TRAUMATIC & NON-TRAUMATIC OCULAR COMPLAINTS IN THE CHILDREN'S EMERGENCY

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TRAUMATIC OCULAR COMPLAINTS IN THE CHILDREN'S EMERGENCY

1. APPROACH TO EYE TRAUMA

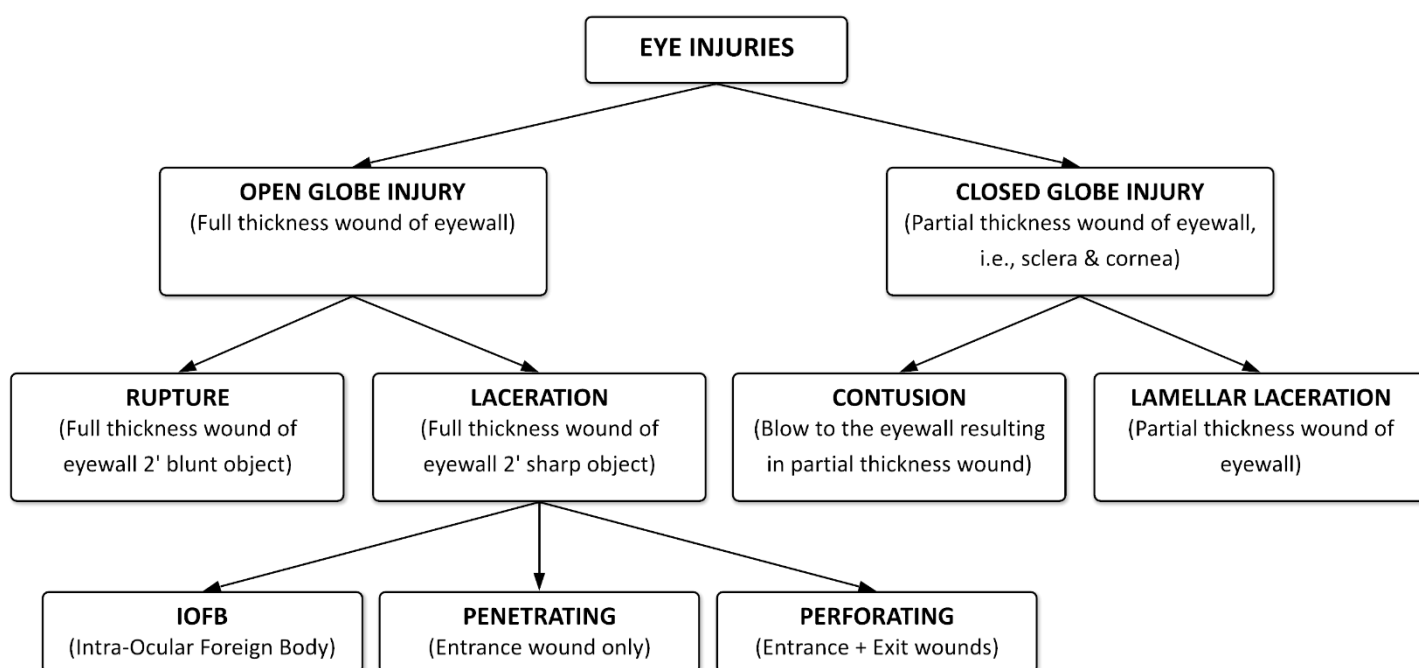


2. INDICATIONS FOR OPHTHALMOLOGY REFERRAL

- All injuries as a result of vegetation or animals
- All ocular infections or injuries in a contact lens wearer
- All chemical exposure with abnormal eye pH
- Patients with significant & persistent ocular pain
- Acutely reduced VA ($\geq 6/12$) after correcting for refraction errors with a pinhole
- Evidence of optic nerve injury or retinal injury (e.g., RAPD on examination)
- All open globe injuries & sharp-force injuries

3. TERMINOLOGY

BIRMINGHAM EYE TRAUMA TERMINOLOGY SYSTEM (BETTS)



4. ASSESSMENT OF OCULAR TRAUMA

4.1 VISUAL ACUITY (VA)

- Patient's best corrected VA should be ascertained (compared with pinhole glasses off/on)
- Arranged in order of decreasing VA:
 - Snellen chart (6/6 → ... → 6/60)
 - Counting fingers
 - Hand movement
 - Light perception
 - No perception to light

4.1.1 SNELLEN CHART¹

- Patients should be assessed with their contact lenses or glasses on
- In patients with previously normal vision, a reduced VA of 6/12 or worse is significant
- Pinhole glasses
 - Correct VA 2' to refractive errors (e.g., astigmatism, myopia) to 6/9 or better
 - If VA (with pinhole glasses) remain 6/12 or worse, an ophthalmology consult should be obtained to evaluate for concurrent pathology other than refractive errors

4.2 HISTORY

- A focused history can help to identify serious eye injuries as physical examination may be limited by the child's ability to cooperate
- Mechanism of Injury

▫ High velocity projectiles ▫ Sticks/organic material ▫ Scissors, sharp glass ▫ Blunt trauma (ball/fists/bats) ▫ Chemical injury²	Intra-ocular/orbital & intracranial FBs ↑ risk of infection Well-defined penetrating injuries Orbital floor fractures, open globe rupture Alkaline vs. Acidic Eye irrigation should be performed ASAP
--	--
- Time of injury
- Ocular health (Baseline vision, contact lens wearer, chronic disease with eye involvement)
- Tetanus vaccination status
- Others: Drug allergies / Medical history / Last meal

¹ Interpretation (e.g., VA 6/9) – The patient can read from 6 meters, what someone with perfect eyesight can read from 9 meters (i.e., patient standing closer to the object than someone with perfect vision for the same clarity)

² Chemicals may not always come in liquid form – solid substances can dissolve/interact, forming solutions that result in chemical injury

4.3 **PHYSICAL EXAMINATION**

- Consider topical anesthesia with 0.5% tetracaine for pain
- Start from the globe and extend outward to the periorbital region.
- Globe
 - Enophthalmos Globe rupture, orbital #
 - Exophthalmos Retrobulbar hemorrhage
 - Foreign body Penetrating/Perforating injury
- Periorbital region
 - Lid laceration Penetrating globe injury, suspect adnexal injury
 - Ecchymosis/bruising Suggestive of underlying pathology
 - Orbital-rim tenderness Orbital wall #
- Light reflexes
 - Hirschberg reflex
 - Pupillary reflexes: RAPD indicates retinal injury or optic nerve injury
- Extra-ocular movements
 - Entrapment of extra-ocular muscle from orbital wall #
 - Commonly inferior margin of the orbit and inability for upward gaze

4.4 **EYE-SPECIFIC BEDSIDE TESTS**

4.4.1 **FLUORESCEIN STAIN**

- Apply a drop of saline or tetracaine to the fluorescein strip
- Gently pull down the patient's lower eyelid and dap the lower lid margin with the fluorescein strip
- Blinking will distribute the fluorescein
- A cobalt blue filter causes fluorescein uptake to appear yellow green, revealing the defects of the cornea

5. **MANAGEMENT OF SPECIFIC CONDITIONS**

5.1 **CHEMICAL EYE INJURY**

- Acidic substances – result in self-limiting burns, forming coagulum which limits spread
- Alkaline substances – results in extensive, penetrative burns as it dissolves tissue, resulting in caustic penetration into the eye, resulting in irreversible blindness
- Normal pH of the eye = 7.0
- Irrigation of the affected eye should begin as soon as possible, prioritized over an eye examination, performing visual acuity, calling ophthalmology etc.
- Contraindication to eye irrigation – open globe injury
- See "[How To](#)" section for details on eye irrigation

5.2 OPEN GLOBE INJURY

- Stop eye examination & apply eye shield to affected eye
- Urgent referral to ophthalmology on-call
- Consider IM tetanus toxoid prophylaxis
- Keep patient fasted (NBM)

5.3 SUBCONJUNCTIVAL HEMORRHAGE

- Rupture of a conjunctival capillary, caused by:
 - Trauma – typically unilateral; commonly 2' minor local trauma
 - Raised pressure from a Valsalva maneuver or cough
- Ophthalmology referral if VA is reduced or following significant sharp- or blunt force trauma to evaluate for associated injury (e.g., Possible IOFB, hyphema, commotio retinae etc.)
- Presentation for minor local trauma: Painless and usually asymptomatic; Some patients might experience a “foreign body” sensation
- Treatment: watchful waiting – resolves within 1 – 2 weeks

5.4 CONJUNCTIVAL FB

- Sweep both superior and inferior fornixes with a sterile Q-tip moistened with sterile saline
- Pull down lower eyelid and perform a sweep from the medial to lateral canthus
- To sweep the superior fornix, the evert the upper eyelid. Steps as follows:
 - Instruct the patient to keep looking downwards
 - Place the tip of a Q-tip at the lid crease (5mm from the lid edge) and apply very light pressure
 - Evert the eyelid over the cotton bud using the eyelashes to gently pull the lid away and upward from the globe
 - Use a new, sterile Q-tip moistened with sterile saline and perform a sweep of the superior fornix from medial to lateral canthus
- Consider eye irrigation if persistently symptomatic
- Look for associated corneal injuries (fluorescein stain)

5.5 HYPHEMA

- Hyphema is defined as blood within the anterior chamber of the eye
- Associated with globe injuries, head injuries & orbital fractures
- ↑ risk of developing glaucoma; Vision-threatening
- Management:

- Urgent referral to ophthalmology
- Eye shield
- Elevate head @ 45 degrees
- Activity restriction
- Avoid NSAIDs

GRADE	ANTERIOR CHAMBER FILLING	VISUAL PROGNOSIS
0	Microhyphemas	75 – 90 %
1	< 33%	75 – 90%
2	33 – 50%	65 – 70%
3	> 50%	25 – 50%
4	100%	25 – 50%

5.6 CORNEAL ABRASION

- Often a result of minor trauma (e.g., eye rubbing, contact with debris etc.)
- Most corneal abrasions heal within 24 to 72 hours
- Pseudomonas keratitis is the most common bacterial keratitis amongst contact lens wearers
- Presentation:
 - Pain, tearing, photophobia, conjunctival injection, reduced VA
 - Foreign body sensation
 - Epithelial defect seen with fluorescein staining with a blue light
- Management includes:
 - Assessing for presence of conjunctival FB
 - Pain relief
 - Preventing secondary bacterial keratitis with topical antibiotics
 - Preventing further injury with topical lubricants
- Referral to ophthalmology if:
 - Patient is a contact lens wearer
 - VA is reduced ($\geq 6/12$)
 - Large defect ≥ 10 mm
 - White lesion is seen on the cornea
- Discharge medication

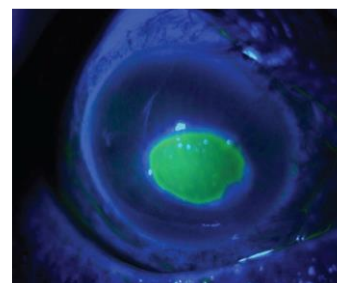


Figure 1 - Fluorescein stain revealing an epithelial defect from a corneal abrasion

Class	Route	Drug	Dose	Frequency
Analgesia	PO	ibuprofen	10 mg/kg (400 mg)	TDS
Antibiotics	Top	Tobramycin eyedrop	1 drop 1 drop	BD – QDS 4H (severe infection)
	Top	Fucithalmic ointment	1 application	BD
	Top	Chloramphenicol eyedrop	1 drop	TDS – QDS
	Top	Chloramphenicol ointment	1 application	BD
Lubricant	Top	Tears Naturale (p/f)	1 – 2 drops	QDS

- Follow up with Ophthalmology in 2 – 3 days

5.7 **ORBITAL FRACTURES**

- Presentation:
 - Periorbital swelling & bruising, orbital rim tenderness or step deformity, enophthalmos
 - Painful or restricted extra-ocular movement with diplopia (classically limitation of upward gaze 2' inferior margin fracture)
- Complications:
 - Orbital compartment syndrome
 - Vagal symptoms associated with extraocular muscle entrapment (nausea, reflex bradycardia)
- Imaging:
 - X-ray orbit:
 - Tear drop sign – herniation of fat/soft tissue through the inferior orbital margin
 - Black eyebrow sign – intra-orbital air
 - CT face: evaluate extent of injury and complications like entrapment of the extra-ocular muscles, as well as full extent of facial injury
- Management:
 - Ophthalmology referral
 - Keep NBM
 - Symptomatic treatment – pain, nausea



Figure 3 – Black eyebrow sign of the left orbit



Figure 3 - Teardrop sign of the left eye

NON-TRAUMATIC OCULAR COMPLAINTS

6. ETIOLOGIES BASED ON PRESENTING SYMPTOMS

PRESENTING SYMPTOM	ETIOLOGY	
	OCULAR	NON-OCULAR
Red eye	Keratitis/Corneal abrasion Episcleritis/scleritis Subconjunctival hemorrhage Conjunctivitis (infectious, allergic, irritant) Hyphema Anterior uveitis	Systemic infection (e.g., <i>leptospirosis</i> , <i>measles</i>) Mucositis (<i>SJS/TEN</i> , <i>Kawasaki disease</i> , <i>MIS-C</i>)
Eye swelling	Chalazion/Hordeolum Dacryocystocele Periorbital cellulitis Orbital cellulitis	Angioedema (e.g., <i>anaphylaxis</i> , <i>allergic reaction</i>) Fluid overload state (e.g., <i>nephrotic syndrome</i> , <i>heart failure</i>) Acute infections (e.g., <i>EBV/CMV IMS</i> , <i>acute sinusitis</i>) Tumors (e.g., <i>rhabdomyosarcoma</i>)
Vision loss / Vision changes	Corneal abrasion Hyphema Glaucoma Uveitis Posterior scleritis Cataracts Retinal detachment Tumors (e.g., <i>retinoblastoma</i>) Optic neuritis	Migraine with visual aura Illicit drug use with mydriasis
Eye pain	Orbital cellulitis Corneal abrasion Infective Keratitis Conjunctival foreign body Chemical injury Glaucoma Trauma	Thrombosis

7. APPROACH TO THE ACUTE RED EYE

- Commonly secondary to:
 - Conjunctivitis (viral, bacterial, allergic, or chemical)
 - Corneal abrasion
 - Conjunctival FB
 - Subconjunctival hemorrhage
- Uncommon causes: glaucoma, episcleritis/scleritis, uveitis
- Discharging non-red eye in infants is likely due to nasolacrimal duct obstruction (NLDO)

RED EYE WITH...	DIFFERENTIALS
Pain, photophobia, discharge	Foreign body Corneal abrasion Keratitis Acute glaucoma
Purulent discharge	Infective conjunctivitis
Itchiness, eyelid swelling & erythema, watery discharge	Allergic conjunctivitis
Dull aching pain	Iritis, episcleritis, scleritis
Subconjunctival hemorrhage	Trauma (need to rule out globe injury) Vigorous coughing / vomiting Valsalva maneuver

8. MANAGEMENT OF SPECIFIC CONDITIONS

8.1 PERIORBITAL CELLULITIS

- Pathogens = Staphylococcus aureus and Streptococcus pyogenes
- Minor local trauma resulting in skin & soft tissue infection
- Presentation:
 - Periorbital erythema, oedema, & tenderness
 - **NO** ocular manifestations or systemic manifestations (e.g., fever)
- Treatment:
 - PO Augmentin 22.5 mg/kg (625 mg) BD x 5 – 7 days
 - Treat underlying cause e.g., chalazion, sinusitis etc.
- Return advice:
 - Ocular manifestations (conjunctivitis, reduced VA, painful/limited eye movements)
 - Fever with progression of periorbital signs
- Follow up with Ophthalmology in 2 – 3 days
 - Not always necessary for older patients who understand red flags & return advice
 - Symptoms should start to improve after 48 hours of effective antibiotics
 - Persist with compresses to the eye

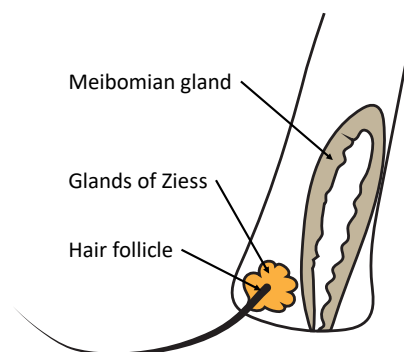
8.2 ORBITAL CELLULITIS

- Requires emergent ophthalmology & ENT consult
- Pathogens = *S. aureus*, *Strep pyogenes*, *Strep pneumoniae*, *H. influenzae*
- Most commonly a direct extension of sinus infection
- Presentation:
 - Periorbital signs (as per periorbital cellulitis)
 - Ocular manifestations: Conjunctival chemosis & injection, painful extraocular movement, limited eye movement, proptosis, reduced VA, optic neuropathy (RAPD)
 - Systemic: Fever is present in up to ⅓ of patients
- Complications include orbital abscess, subperiosteal abscess, optic neuropathy, meningitis, cavernous sinus thrombosis
- Investigations:
 - CT orbits for confirmation of diagnosis and assessment of complications that might require surgical drainage.
- Choice of antibiotics:
 - IV Augmentin 30 mg/kg (1200 mg) TDS

8.3 EYELID PATHOLOGY

8.3.1 HORDEOLA & CHALAZIA

- Hordeolum (stye) an acute infection of an oil gland
 - External: glands of Zeiss within the hair follicle
 - Internal: Meibomian gland posterior to the eyelashes
 - Tend to be more erythematous & painful
- Chalazion
 - Blockage of the Meibomian gland → inciting an inflammatory response which can become chronic & granulomatous; Often begin as a hordeolum
 - Less painful
- Treatment
 - Both are self-limiting, oral antibiotics are not typically indicated
 - First-line treatment = Warm compress with lid hygiene (2 – 3 times/day); it can take up to 1 week for lesions to improve
 - This is often inadequately performed by caregivers (Refer to “[Warm Compress](#)” for the appropriate advice for caregivers)
 - Adjunct treatment: Topical fucithalmic BD
 - Lesions that do not improve with 1st line treatment would warrant a referral to Ophthalmology for incision & drainage



8.3.2 BLEPHARITIS

- Inflammation of the lid margin commonly secondary to bacterial colonization
- Presentation:
 - Flaking & crusting at the lash base
 - May be associated with corneal irritation, reactive conjunctivitis, or chalazia
- Treatment with warm compresses and lid margin hygiene

8.3.3 DACRYOADENITIS (LACRIMAL GLAND)

- Uncommon in pediatrics
- Erythema & swelling over the supero-temporal orbit / upper eyelid
- Multiple etiologies – idiopathic, viral/bacterial, inflammatory
- Most common etiology = idiopathic (self-limiting)
- May treat with NSAIDS

8.3.4 NASO-LACRIMAL DUCT OBSTRUCTION (NLDO)

- Most present within the 1st month of life, though some present later as tear production is only fully developed at 2 months of life.
- 80 – 90% spontaneously resolve by the age of 6 – 10 months
- Presentation:
 - Intermittent tearing & debris over unilateral and/or bilateral eyes
 - Crusting & some mucopurulent discharge is 2^o to bacterial overgrowth from stagnation of tears (rather than infection); Conjunctival injection is not common and other etiologies should be entertained
- Diagnosis can be confirmed with the (fluorescein) “dye disappearance test” (see “How To” section)
- Management
 - Lid hygiene with boiled clean water and cotton balls (see below)
 - Topical antibiotics for bacterial overgrowth
 - Crigler’s massage (see “How To” section)

8.3.5 DACRYOCYSTOCELE

- Forms as a result of NLDO distally
- Presents as a bluish swelling of the skin over nasolacrimal sac inferior to the medial canthus with superior displacement of the medial canthal tendon
- Ophthalmology consult should be obtained as there is significant risk of infection, dacryocystitis and nasal obstruction resulting in respiratory compromise



8.3.6 DACRYOCYSTITIS (NASOLACRIMAL DUCT)

- Sudden onset of erythema & pain at the medial canthal region
- Can be associated with local cellulitis of the overlying skin or result in invasive infections like orbital cellulitis, meningitis, and sepsis
- Management:
 - Acute referral to ophthalmology
 - If no evidence of invasive infection, can be treated with oral and topical antibiotics with a close outpatient review
 - PO Augmentin 22.5 mg/kg (625 mg) BD for 7 – 10 days, and
 - G Tobramycin drops QDS
 - Consider a reverse Crigler's massage (massage in reverse)
 - Expectant management – symptoms might worsen over 24 – 48H before getting better with PO antibiotics
 - Follow up with Ophthalmology in 2 – 3 days

8.4 CONJUNCTIVITIS

- Refers to conjunctival inflammation with dilatation of vessels, hyperemia and occasionally discharge

8.4.1 NEONATAL CONJUNCTIVITIS (OPHTHALMIA NEONATORUM)

- Conjunctivitis in patients < 1 month old
- Vertical transmission during passage through the mother's genital tract at birth
- Referral to ophthalmology for microbiological sampling & diagnosis

8.4.1.1 NEISSERIA GONORRHOEAE

- Presents in the 1st week of life with copious mucopurulent secretions & severe conjunctival injection
- Complications:
 - Local – corneal ulcerations, corneal perforation
 - Systemic – disseminated infection (bacteremia, meningitis, & arthritis)
- Ocular emergency & requires review by ophthalmology
- Patient should be admitted septic workup & IV ceftriaxone
- Management:
 - Ophthalmology referral for microbiological samples
 - Admit patient for septic evaluation and observing for response to therapy
 - Treatment: IM or IV ceftriaxone 50 mg/kg (125 mg) once

8.4.1.2 CHLAMYDIA TRACHOMATIS

- Presents in the 2nd week of life with mild conjunctivitis & white, stringy discharge which can become mucopurulent, marked eyelid swelling with chemosis
- Complications: pseudo-membrane formation
- Good prognosis without complications if treated
- Management:
 - Ophthalmology referral for microbiological samples
 - Treatment: PO erythromycin 12.5 mg/kg (250 mg) q6H for 14 days
 - O&G referral for mother → screening & treatment of STIs (can affect fertility if left undiagnosed & untreated)
- Associated with afebrile pneumonia of infancy (chlamydia pneumonitis)
 - Presents at 4 – 6 weeks of life with cough, nasal discharge, tachypnea, increased work of breathing with bilateral crepitations on auscultation with mild to moderate hypoxia
 - Peripheral eosinophilia on FBC

8.4.1.3 HERPES SIMPLEX VIRUS (HSV) CONJUNCTIVITIS

- Presents within the 1st 6 weeks of life (most commonly within the 1st 2 weeks)
- Neonatal HSV infection has 3 modes of presentation:
 - Skin, eye & mouth (SEM)
 - CNS disease (can occur in isolation without SEM or disseminated disease)
 - Disseminated disease (sepsis like with multi-organ involvement: liver, lung, CNS, heart, renal, GI, bone marrow, hematologic)
- Presentation: conjunctival injection, periorbital vesicular lesions
- Complications: keratoconjunctivitis, chorioretinitis with permanent vision impairment
- Patient should be admitted for IV acyclovir 20 mg/kg q8H for 14 days (SEM presentation)

8.4.2 CHILDHOOD CONJUNCTIVITIS**8.4.2.1 VIRAL CONJUNCTIVITIS**

- Viral conjunctivitis is more common than bacterial conjunctivitis
- Symptoms:
 - Bilateral conjunctival injection (unilateral presentation does not exclude a viral pathology); unilateral symptoms with contralateral involvement after 1 – 2 days
 - Watery, non-purulent discharge
- Keratoconjunctivitis
 - Conjunctivitis with corneal involvement
 - Presentation: FB sensation with refusal to open eye, reduced VA, corneal opacities, photophobia
 - Ophthalmology referral indicated
- Treatment is mainly symptomatic: topical lubricants & antihistamines

8.4.2.2 BACTERIAL CONJUNCTIVITIS

- Pathogens: Streptococcus pneumoniae, Haemophilus influenzae, Moraxella catarrhalis, Staphylococcus aureus
- Suspect Pseudomonas aeruginosa in contact lens wearers – urgent ophthalmology consult indicated → can result in corneal perforation within 24H
- Symptoms:
 - Unilateral conjunctival injection
 - Thick, purulent discharge → eye stuck shut in the morning
 - Discomfort, chemosis
- Treatment with topical antibiotics & lubricants
- Hyperacute presentation (N. gonorrhea conjunctivitis)
 - If chlamydia has not been excluded, to give as part of an appropriate combination regimen
 - Presentation:
 - Rapid onset (within 12H of inoculation)
 - Profuse purulent discharge with erythema, irritation & tenderness to palpation, chemosis, lid swelling, pre-auricular lymphadenopathy
 - Vision threatening and warrants an urgent ophthalmology consult
 - Treatment with systemic antibiotics:
 - Age ≥ 3 months (< 45 kg): IM ceftriaxone 25 – 50 mg/kg (250 mg) once
 - Age ≥ 3 months (≥ 45 kg): IM ceftriaxone 25 – 50 mg/kg (500 mg) once
 - Adolescents: IM ceftriaxone 1000 mg once

8.4.2.3 ALLERGIC CONJUNCTIVITIS

- Symptoms:
 - Itchy eyes – can present with frequent blinking (in the absence of ocular itching, other differentials should be entertained)
 - Conjunctival erythema, eyelid edema, watery, non-purulent discharge
- Associated with other atopic diseases
- Management:
 - Trigger avoidance – HDM, pet dander etc. (the absence of triggers does not exclude this diagnosis)
 - Medication:
 - (≥ 3 years) Topical olopatadine (Patanol) 1 drop BD
 - (≥ 3 years) Topical ketotifen (Zaditen) 1 drop BD
 - Tears Naturale (Preservative Free) 1 drop QDS
 - Systemic antihistamines
 - Cold compresses
 - Follow up with Ophthalmology in 1 – 2 weeks

8.5 GLAUCOMA

- Progressive optic neuropathy often with raised intra-ocular pressure with peripheral visual field deficits → central deficit → complete blindness

PRIMARY INFANTILE GLAUCOMA	JUVENILE GLAUCOMA
Presents by 2 – 3 years of age	Presents after 3 years of age (≥ 4 years) AD inheritance with a significant family history
Classic triad of chronic intermittent epiphora, photophobia, and blepharospasm Signs related with $\uparrow$ IOP: corneal edema with clouding & ocular enlargement (buphthalmos)	Insidious onset of raised IOP → minimal symptoms; no classic signs of buphthalmos or corneal edema Often already has significant vision issues @ presentation

- An urgent ophthalmology consult should be obtained if suspected

9. HOW TO...

- General rules of thumb
 - Apply eye drops first
 - Wait 5 minutes between different eye drops
 - Wait 5 minutes between eye drop and eye ointment
 - Wait 10 minutes between different eye ointments

9.1 APPLY EYE DROPS

- Wash hands thoroughly with soap & water. Dry hands before applying the eyedrop
- Shake bottle gently before removing the cap and placing it on a clean tissue paper
- Position your child:
 - If seated, to tilt head backward slightly, and look upwards
 - Lie down if uncomfortable, or unable to tilt head
- Hold the bottle like a pen. Ensure the tip of the bottle does not touch the eye, eyelid, or eyelashes to avoid contamination
- Use a finger to pull down lower eyelid and squeeze bottle to instill one drop into the pocket of the lower eyelid
- Child should close eyes gently
- With eyes still closed, perform punctal occlusion by gently pressing against the inner corners of the eye for 1 – 2 minutes
- Recap bottle and store in a cool, dry area (or as instructed on the bottle)

9.2 APPLY EYE OINTMENT

1. Wash hands thoroughly with soap & water. Dry hands before applying the ointment
2. Remove the cap from the tube and place on a clean tissue paper
3. Tilt head backwards slightly, look up, and pull down the lower eyelid
4. Hold the tube like a pen. Ensure the tip of the tube does not touch the eye, eyelid, or eyelashes to avoid contamination
5. Squeeze 0.5 – 1cm of ointment into the pocket of the lower eyelid
6. Blink gently a few times to spread the ointment then keep eyes closed for 1 – 2 minutes
7. Wipe excess ointment from the eyelid/eyelashes with a clean tissue paper
8. Recap the ointment tube

9.3 PERFORM WARM COMPRESS

1. Soak a clean towel in water that is very warm to hot but not scalding
 - a. Water temperature must be hot enough to liquefy the oily secretions from the oil glands in the eyelids
 - b. Caregivers should perform the first warm compress on themselves to test the water temperature before applying the warm compress on their child
2. Wring out excess water but not until it is completely dry (this is so that enough heat is retained in the towel for the warm compress to be effective)
3. Place the wet towel against closed eyelids and lightly massage for about 20 to 30 seconds, or until the towel is cool
4. Repeat this process for 5 minutes in total, for at least 2 – 3 times a day, until the lesion is completely resolved

9.4 PERFORM LID HYGIENE

1. Wash hands with soap & water
2. Fold cotton linen into a small rectangle & apply eyelid cleanser/boiled clean water on linen
3. Close eyes gently; Eyelid margins will not be prominent if eyes are squeezed
4. Gently lift upper eyelid with a finger and wipe the margin where the eye lash meets the skin several times with the cotton linen
5. Repeat with lower eyelid
6. Wipe away eyelid cleanser with a damp towel especially at the eye lash region

9.5 PERFORM DYE DISAPPEARANCE TEST

(FYI, but not necessary for the diagnosis of NLDO)

1. Place a drop of topical anesthetic to the eye
2. Place a drop of fluorescein-stained saline into the inferior fornix of each of the patient's eyes
3. Wipe away excess tears from the eyelids
4. Observe for 5 minutes – do not allow rubbing of eyes; do not wipe away tears that drain into the cheek
5. Dye should completely disappear from eyes after 5 minutes. Test is abnormal if a large amount of dye persists or has drained over the lower eyelid onto the cheek

9.6 PERFORM CRIGLER'S MASSAGE

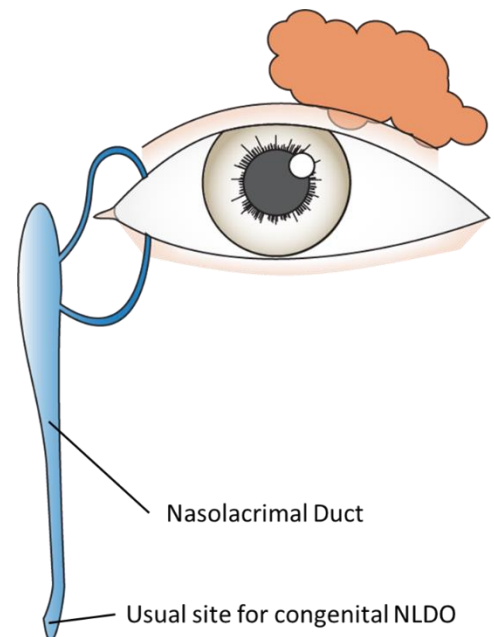
Crigler's maneuver creates high intraluminal hydrostatic pressure within the nasolacrimal sac, causing it to give way at its weakest point (site of the nasal opening). This is achieved through occluding the proximal sac and massaging down its length distally, preventing reflux of fluid through the proximal opening of the sac.

9.6.1 **CONTRAINDICATIONS**

- Acute dacryocystitis
- Obstruction 2' canaliculi scarring or punctal agenesis

9.6.2 **STEPS**

1. This is best performed after the child has fed and lying in the caregiver's lap
2. Lubricant eyedrops or baby oil can be used as lubricant
3. The caregiver should place their ipsilateral thumb or index finger over the medial canthus to occlude the common canaliculus to block the exit of material through the lacrimal punctum.
4. Stroke downward to increase hydrostatic pressure within the nasolacrimal sac.
5. Perform 5 – 10 strokes approximately 4 times a day (every other q3H feed)



9.7 PERFORM EYE IRRIGATION

Indications:	- Removal of foreign bodies from the eye - Treatment of chemical injuries	
Contraindications:	Open globe injury	
Equipment:	- Litmus paper 0.5% tetracaine eye drops - Normal Saline (0.9% NaCl) IV bag	- IV infusion set - Kidney dish (non-sterile) - IncoPad (incontinence pad)
STEPS	DETAILS	
1. Assess eye pH	- Pull down lower eyelid to test inferior fornix with litmus paper - If pH ≠ 7.0, begin eye irrigation (If applied first, note that tetracaine has an acidic pH)	
2. Apply anesthesia	- Apply 1 – 2 drops of 0.5% tetracaine to the inferior fornix - Effect lasts approximately 30 minutes; consider repeating topical anesthetic every 30 minutes if required (based on patient feedback)	
3. Prepare patient	- Patient should lie supine with head turned slightly to the side of the affected eye - Place kidney dish against the ipsilateral cheek to catch run off - Place incopads so that patient does not get wet	
4. Prepare equipment	- Put on PPE (especially eye shield & protective gown in case of splash) - Connect IV infusion set to the bag of normal saline - Adjust clamp to increase flow rate enough to achieve a steady stream that does not splash when held 3 to 4 cm from the patient's eye	
5. High-volume, low-pressure irrigation (at least 2 litres)	- Start irrigation from the medial canthus to the lateral canthus of each fornix, alternating between the inferior and superior fornixes - Retract the eyelid to expose as much of the fornix as possible - Irrigate continuously for 30 minutes before performing the 1st post-irrigation pH assessment	
6. pH surveillance	- Continue irrigation if the eye pH is still abnormal - Eye pH can be surveyed every 15 – 30 minutes - Stop once a neutral pH is achieved - The eye pH should be reassessed <u>5 min</u> & <u>30 min</u> after irrigation was stopped to assess for rebound from pockets of chemical in the fornixes	

DRUG DOSES

ANNEX A:

DRUG	DOSE	FREQUENCY	REMARKS
ANESTHETIC			
Tetracaine 0.5%	1 drop (5 drops)	q3 – 5 mins	Titrated to effect Advise against touching eye for 20 mins after instillation due to potential for accidental injury from poor sensitivity pH 4.54 (acidic) – to take into consideration when testing eye pH
ANTIBIOTICS			
EYEDROPS			
Tobramycin 3%	1 drop	BD – QDS	Suitable for age ≥ 2-months For severe infections (e.g., pseudomonas keratitis), max. 2 drops every hour until improvement
	1 drop	q4H	Severe infection up to 2 drops per hour
Chloramphenicol 0.5%	1 drop	TDS – QDS	May cause local hypersensitivity reactions Suitable for neonates
Levofloxacin 1.5%	1 drop	QDS	≥ 1-years
OINTMENTS			
Fucithalmic 1%	1 application	BD	7-day course
Chloramphenicol 1%	1 application	QDS	
Chlortetracycline 1%	1 application	BD	Sandy sensation in the eye, may not be tolerated by younger patients
ANTI-HISTAMINE			
Olopatadine 0.1% (Patanol)	1 drop	BD – QDS	≥ 3-years
Ketotifen 0.025% (Zaditen)	1 drop	BD	≥ 3-years
LUBRICANT			
Tears Naturale (preservative free)	1 drop	QDS-PRN	
Carbomer Eye Gel (Vidisic)	1 application	QDS-PRN	
Duratears Eye Ointment	1 application	QDS-PRN	